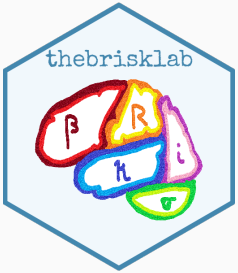
Motion artifacts, selection bias, and solutions to improve pediatric brain connectivity studies



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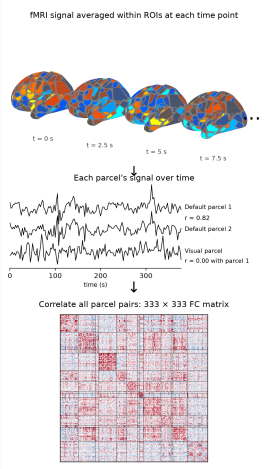
Unprocessed resting-state fMRI from a median-motion autistic boy



https://youtu.be/OuNCONMSfaQ

Functional connectivity

- Resting-state fMRI: blood-oxygen-level-dependent (BOLD) signal collected every 1–2.5 s while the participant rests.
- Average the signal within regions of a brain parcellation.
- **Functional connectivity (FC)** = correlations between regional time series.
- FC is used to study brain organization and its differences in neurodevelopmental disorders.



Three theories of autism — all built on resting-state fMRI

- **Connectivity theory**: hypoconnectivity and hyperconnectivity between brain regions (Di Martino et al., 2013; Ilioska et al. 2023).
- **Triple network theory**: alterations between the default mode, salience, and central executive networks (Menon 2011; Abbott et al. 2016; Nair et al. 2026).
- **Idiosyncratic theory**: greater variability between autistic individuals than between non-autistic individuals (Hahamy et al. 2015).

- All three are based on correlations between fMRI signals across time — exactly what motion distorts.
- Motion produces patterns resembling reported autism differences (Deen and Pelphrey, 2012).

Three data sources in this talk

ABIDE I/II

Autism Brain Imaging Data Exchange

377 children (132 autistic)

ages 8–13

KKI + NYU sites; TR 2–2.5 s

Methods applications:

MoCo for autism differences, causal inference for stimulant impacts

ABCD

Adolescent Brain Cognitive Development

7,393 children (545 ADHD), ages 9–10

consortium preprocessed

Selection bias at scale;

MoCo applied to ADHD

Emory Autism Brain Connectivity Study

166 children (86 autistic), ages 8–13

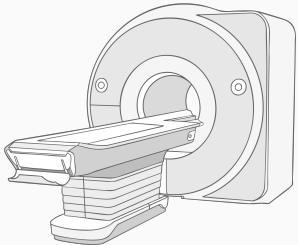
4 rs-fMRI runs; TR 1.127 s

with Dr. Sarah Shultz

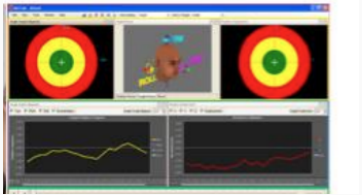
Selection bias in a new cohort; future connectivity analysis

Emory Autism Brain Connectivity Study

- 8–13 years old: 86 autistic and 80 non-autistic.
- Mock training with head-motion feedback (MoTrak).
- Collect T1w, 4 rs-fMRI, 2 task fMRI with movie watching and eye-tracking.



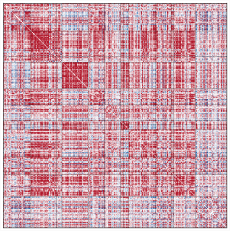


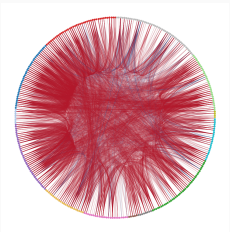


Motion artifacts and quality control

Example single subject functional connectivity

Unprocessed

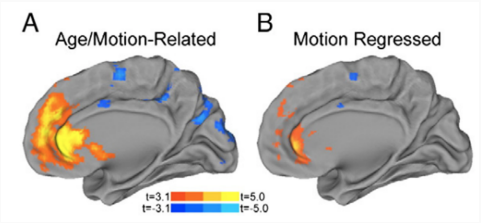




Top: Fisher-z correlation matrices (shared scale). Bottom: edges with $|z| \geq 0.5$; red positive, blue negative.

Motion incorrectly interpreted as development

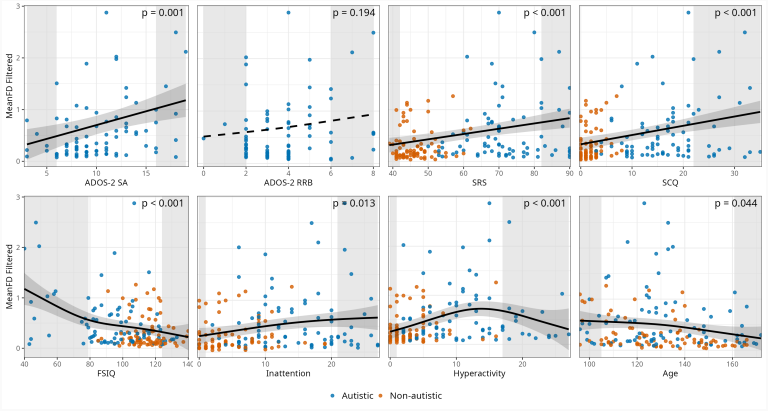
- Early landmark finding: default mode connectivity matures from childhood to adulthood (Fair et al., 2008, *PNAS* — cited 1,720 times).
- Younger children move more (Satterthwaite et al., 2012).
- Age-related posterior–anterior connectivity increases were largely motion artifacts.



"It really, really, really sucks. My favorite result of the last five years is an artifact."

— *Steven Petersen, Washington University in St. Louis*

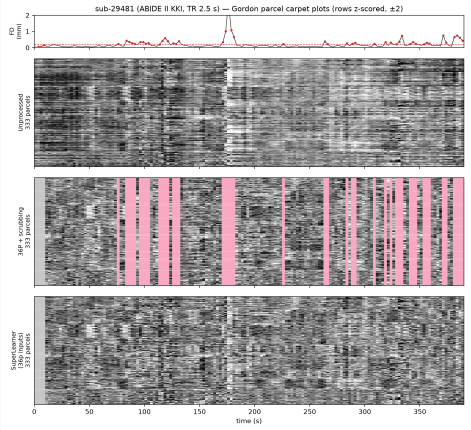
Emory study: motion increases with autism symptoms ($Z \to M$)



GAMs of mean filtered FD on participant characteristics, adjusting for age and sex; 163 children.

Standard preprocessing: nuisance regression + scrubbing

- Rigid-body motion correction aligns volumes over time.
- **Scrubbing**: remove frames with framewise displacement (FD) > 0.2 mm (Power et al., 2014).
- **36P nuisance regression**: motion parameters, global signal, WM, CSF, plus derivatives and quadratics (Ciric et al., 2017).
- Motion patterns are complex; regression alone is inadequate (Power et al., 2014).

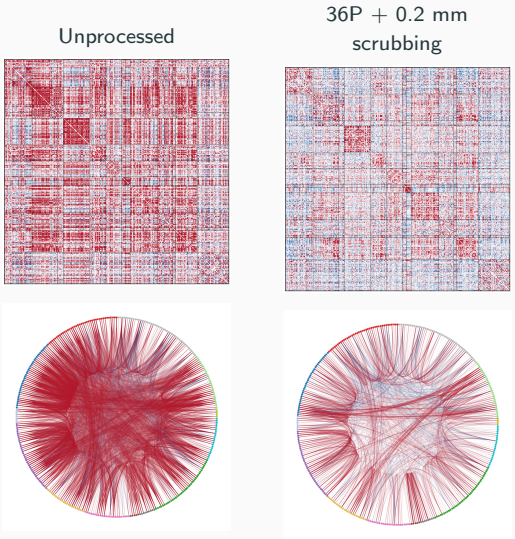


36P nuisance regression + 0.2 mm scrubbing



https://youtu.be/10ce1EQRsJk

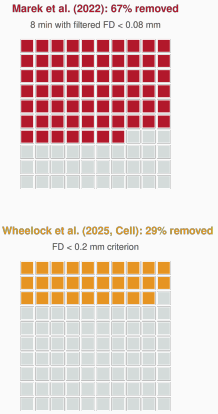
Example single subject functional connectivity



Top: Fisher-z correlation matrices (shared scale). Bottom: edges with $|z| \geq 0.5$; red positive, blue negative.

Motion quality control: the rules of the game

- Current practice: scrub frames with FD > 0.2 mm, then **remove the participant** if < 5 minutes of data remain (Power et al., 2014).
- Stricter criteria are increasingly used: > 375 frames with FD < 0.2 mm (Hagler Jr et al., 2019), mean FD < 0.2 mm, 8 minutes at FD < 0.08 mm (Marek et al., 2022).
- Large reductions in sample size n: a reproducibility study removed 67% of $\sim$11,900 children (Marek et al., 2022).

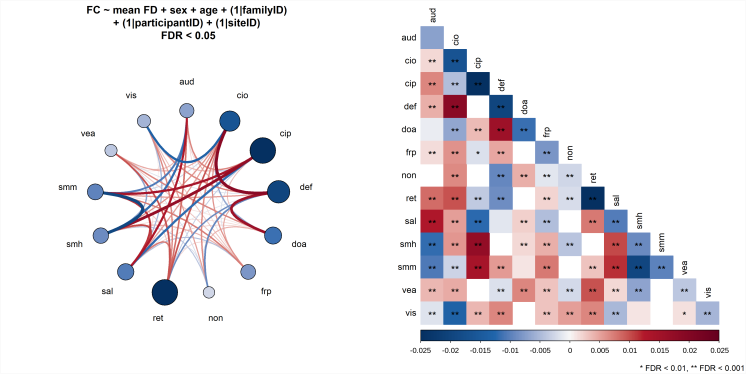


Emory ABC: Quality control removes the majority of autistic children

Criterion	**Autistic** ($n = 86$)	**Non-autistic** ($n = 80$)	p
5 mins, <0.2 mm	58 (67.4%)	66 (82.5%)	0.032
>375 frames	31 (36.0%)	54 (67.5%)	<0.001
mean FD < 0.2 mm	22 (25.6%)	44 (55.0%)	<0.001
8 mins, <0.08 mm	5 (5.8%)	18 (22.5%)	0.003

Data retention under four motion QC inclusion criteria, Emory Autism Brain Connectivity Study. Criteria from Power et al. (2014), Hagler Jr et al. (2019), Shafiei et al. (2025), Marek et al. (2022).

ABCD: Preprocessing does not remove the motion–FC relationship



ABCD 7.0 consortium-processed functional connectivity, computed *after* censoring high-motion frames:

FC remains strongly associated with mean framewise displacement across children.

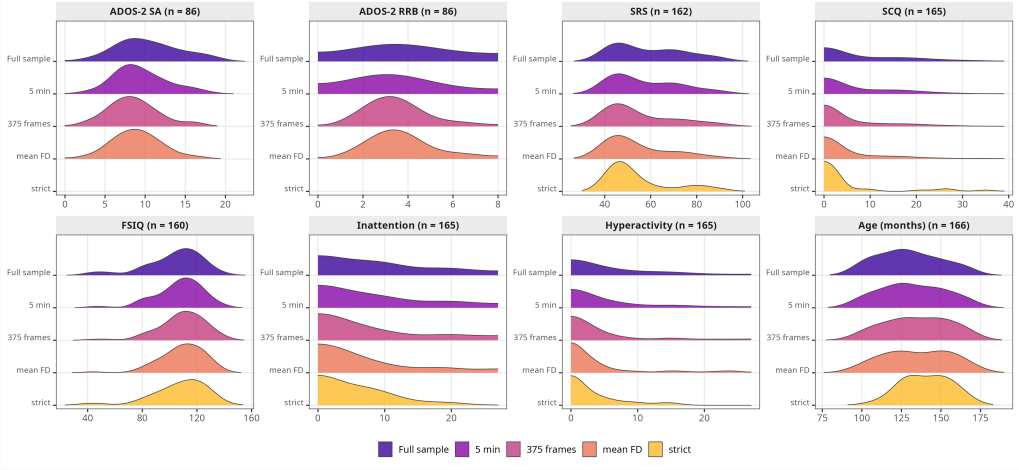
Two problems

1. **Residual confounding:** even after nuisance regression and scrubbing, functional connectivity remains related to motion.
2. **Selection bias:** the children removed by quality control are not a random sample.

The rest of this talk: evidence for problem 2, then methods that address both.

Motion quality control creates selection bias

Emory study: stricter criteria shift the analyzed sample



Densities of participant characteristics in the full sample and in the sample retained by each criterion. ADOS-2 panels: autistic children only; the strict criterion retains 5 autistic children, so its ADOS-2 ridges are omitted.

Same story in data from an ABIDE site (Kennedy Krieger Institute)

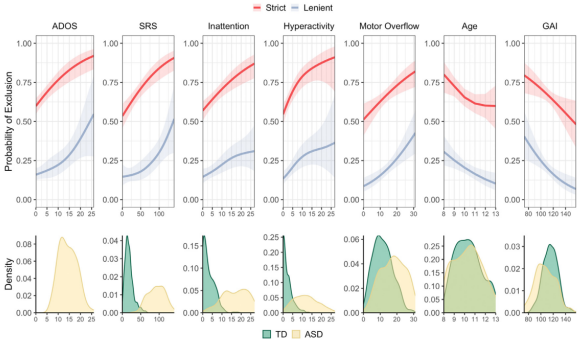


Fig. 4. rs-fMRI exclusion probability changes with phenotype and age. Univariate analysis of rs-fMRI exclusion probability as a function of participant characteristics. From left to right: Autism Diagnostic Observation Schedule (ADOS) total scores, social responsiveness scale (SRS) scores, inattentive symptoms, hyperactive/impulsive symptoms, total motor overflow, age, and general ability index (GAI) using the lenient (slate blue lines, all FDR-adjusted $p < 0.01$), and strict (red lines) motion quality control (all FDR-adjusted $p < 0.03$). Variable distributions for each diagnosis group (included and excluded scans) are displayed across the bottom panel (TD=typically developing, green; ASD=autism spectrum disorder, yellow).

Nebel, Lidstone, Wang, Benkeser, Mostofsky, Risk (2022), *NeuroImage*.

Selection bias in ABCD: demographics ($X \to \Delta$)

	Non-ADHD ($n = 6{,}848$)			ADHD ($n = 545$)		
Demographic (X)	Excluded	Included	p	Excluded	Included	p
Age, mean (SD)	9.84 (0.59)	9.97 (0.62)	<0.001	9.92 (0.63)	9.96 (0.63)	0.423
Male	820 (58.7%)	2577 (47.3%)	<0.001	131 (73.2%)	251 (68.6%)	0.270
Race: Black	233 (16.7%)	652 (12.0%)	<0.001	31 (17.3%)	53 (14.5%)	0.584
Hispanic	311 (22.2%)	1059 (19.4%)	0.019	32 (17.9%)	68 (18.6%)	0.842
Income $<$ 50k	464 (33.2%)	1413 (25.9%)	<0.001	57 (31.8%)	106 (29.0%)	0.493
Parents: graduate degree	443 (31.7%)	2129 (39.1%)	<0.001	52 (29.1%)	118 (32.2%)	0.417

- ABCD 7.0 baseline: $n = 7{,}393$ children with observed FC; motion exclusion removes 1,577 (21.3%).
- Demographic selection occurs **within the non-ADHD group** (younger, male, Black, Hispanic, lower-SES children excluded); within the ADHD group, no demographic variable differs ($p \geq 0.10$).

Selection bias in ABCD: symptoms and cognition ($Z \to \Delta$)

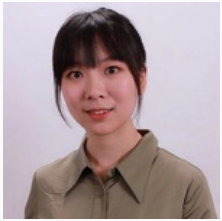
	Non-ADHD ($n = 6{,}848$)			ADHD ($n = 545$)		
Symptom / cognition (Z)	Excluded	Included	p	Excluded	Included	p
Crystallized cognition	50.46 (11.03)	52.15 (11.56)	<0.001	49.35 (11.23)	50.27 (11.37)	0.373
Fluid cognition	44.51 (11.33)	46.90 (11.01)	<0.001	40.88 (11.54)	43.62 (11.32)	0.009
KSADS inattention	0.56 (1.74)	0.32 (1.31)	<0.001	7.26 (1.71)	7.37 (1.64)	0.501
KSADS hyper/impulsive	0.34 (1.15)	0.18 (0.89)	<0.001	5.75 (2.38)	5.30 (2.48)	0.039
CBCL attention problems	53.77 (5.18)	52.81 (4.65)	<0.001	66.64 (8.84)	65.52 (8.80)	0.168
Sleep problems	36.88 (7.61)	35.89 (7.19)	<0.001	44.68 (12.32)	44.40 (11.52)	0.799
Mean FD	0.68 (0.50)	0.18 (0.15)	<0.001	0.86 (0.60)	0.20 (0.15)	<0.001

- Children with ADHD are excluded more often (32.8% vs. 20.4%), but within the ADHD group exclusion does not select on symptom severity.
- Within the non-ADHD group, exclusion removes children with more subclinical symptoms and lower cognition $\Rightarrow$ a "super control" comparison group.

Solution 1: Estimate connectivity under low motion (MoCo)

Motion Control (MoCo) in functional connectivity studies

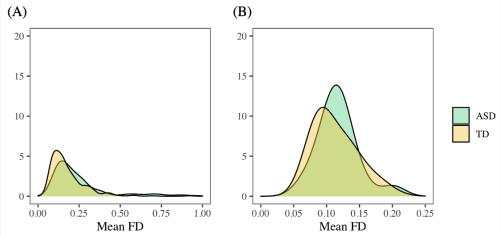
- Ran, J., Shultz, S., Risk, B. B., and Benkeser, D. (2026). Nonparametric motion control in functional connectivity studies in children with autism spectrum disorder. *Biometrics*, 82(2).
- Works on the **processed connectivity** you already have.
- Retains **all** participants — no motion exclusion.



Jialu Ran

Stochastic intervention: what would FC look like under low motion?

- Marginal estimand formulation of a stochastic intervention on a mediator (motion) (Díaz et al., 2021).
- Replace each child's motion M by a draw from the **low-motion distribution** $M \sim P_{M|\Delta=1,A=0,X}$.
- "Tolerable" motion: non-autistic children that pass motion QC.
- Intuition: what the brain looks like under counterfactual low motion.



Motion-Controlled Estimand (MoCo)

The difference between the autistic and non-autistic brain when all children have a tolerable amount of motion.

- A: non-autistic (0), autistic (1); M: mean framewise displacement.
- X: demographic variables; Z: variables related to autism symptoms.

$$\int \Big[\big\{\mu_{Y|A,M,X,Z}(1,m,x,z)\, p_{Z|A,X}(z|1,x) - \mu_{Y|A,M,X,Z}(0,m,x,z)\, p_{Z|A,X}(z|0,x)\big\} p_{M|\Delta=1,A,X}(m|0,x)\, p_X(x)\Big]\, dz\, dm\, dx\ .$$

MoCo: estimation

- One-step estimator.
- Nuisance regressions estimated with **SuperLearner**: random forest, xgboost, multivariate adaptive regression splines, LASSO, ridge, generalized additive models, and GLMs (with and without interactions and forward stepwise selection).
- Conditional density of motion estimated with the highly adaptive lasso (Hejazi et al., 2022).
- Cross-fitting with 5-fold cross-validation.
- Simultaneous confidence bands for FWER control.

MoCo: multiple robustness

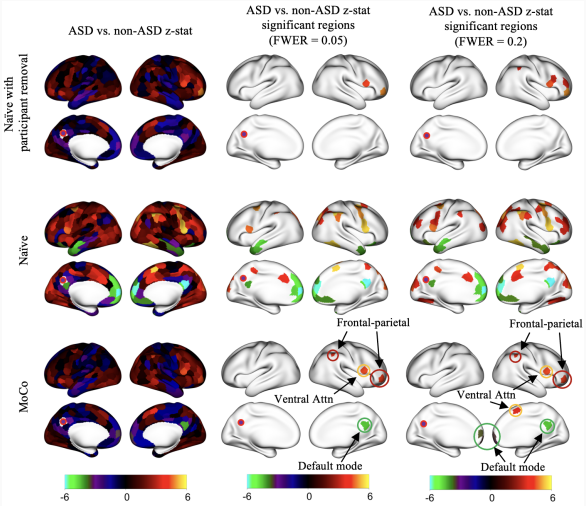
	$\mu_{n,Y\|A,M,X,Z}$	$\eta_{n,\mu\|A,M,X}$	$\xi_{n,a,\eta\|X}$	$\bar{\pi}_{n,0}$	$\pi_{n,a}$	$p_{n,M\|\Delta=1,A,X}$	$p_{n,M\|A,X,Z}$
(B2.1)					✓	✓	✓
(B2.2)			✓			✓	✓
(B2.3)	✓	✓		✓	✓		
(B2.4)	✓				✓	✓	
(B2.5)	✓		✓			✓	

Each row is a setting for consistency: check marks indicate the nuisance parameters which, when they converge to the true functions, give $E[D_{P',a}(O)] = 0$ and $\theta^+_{n,a} \to \theta_a$.

ABIDE application: posterior default mode connectivity

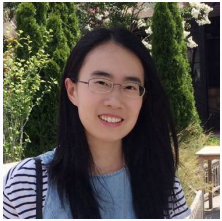
- 377 children (245 non-autistic, 132 autistic), ages 8–13, KKI + NYU.
- A: autism diagnosis; X: age, sex, handedness; Z: ADOS, FIQ, stimulant and non-stimulant medication status; M: mean FD; Δ: passes motion QC; Y: posterior DMN seed connectivity.
- Standard QC (5 min after 0.2 mm scrubbing) would exclude **74% of autistic** and 49% of non-autistic children.
- FC: Schaefer 400, seed = posterior DMN hub; nuisance regression of motion parameters, global signal, WM, CSF; site harmonized with ComBat.

ABIDE inference: naive vs. MoCo



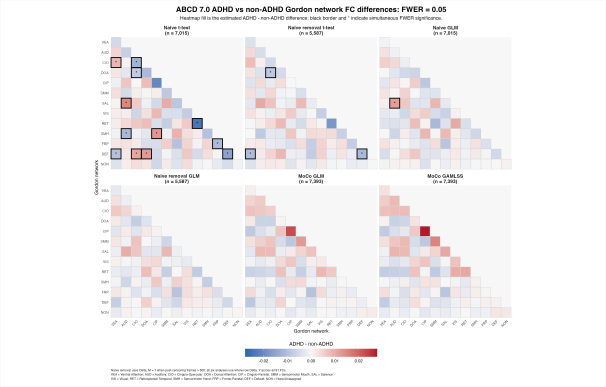
MoCo applied to ADHD in the ABCD dataset

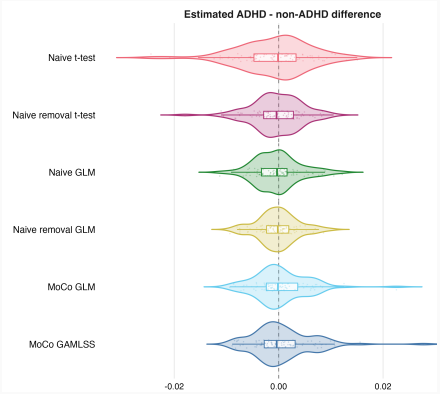
- **Goal**: examine functional connectivity differences between children with and without ADHD ($n = 7{,}393$, ages 9–10; 91 Gordon network pairs).
- **Recall**: in these consortium-processed data, FC is still strongly related to mean FD, children with ADHD move more (mean FD 0.41 vs. 0.29), and motion QC selects a "super control" non-ADHD group.
- **Previous findings**: ADHD-related FC differences have been widely reported, including reduced segregation between the default mode and central executive networks.



Xiyan Tan

ABCD results: ADHD vs. non-ADHD functional connectivity





Estimated ADHD − non-ADHD differences across 91 Gordon network pairs; boxed cells FWER < 0.05.

ABCD interpretation

- The naive t-test finds 12 significant network pairs; adjusting for demographics and mean FD leaves 0–3; MoCo finds 0 while retaining all 7,393 children.
- The naive "ADHD differences" reflect motion artifact and demographic confounding; with motion handled, ADHD–non-ADHD FC differences at ages 9–10 are small.
- Naive removal GLM removes 33% of children with ADHD and appears to still be biased — within-network connectivity signs flip in MoCo.

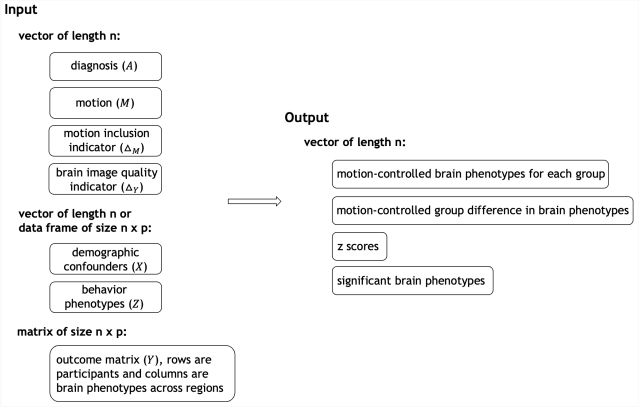
MoCo software

- R package: inputs are diagnosis, motion, inclusion indicator, confounders, phenotypes, and the FC matrix.
- Outputs motion-controlled brain phenotypes, group differences, and z-scores.





github.com/thebrisklab/MoCo



Solution 2: Remove motion from the time series, inside a causal framework

From the outcome level to the time-series level

- MoCo used correlations.
- Can we clean the **time series** better without discarding frames?
- Can we use adaptive machine learners in the time-series processing while retaining $\sqrt{n}$-consistency of the treatment effect? (Derived outcomes)



Zihang Wang

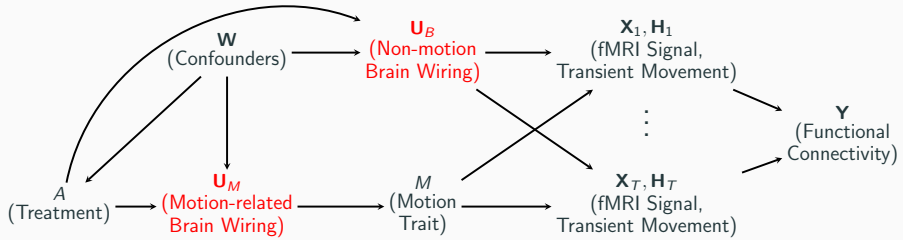
Wang, Z., Nabi, R., & Risk, B. B. (2026).

arXiv:2602.02240

Scientific question: stimulant effects on connectivity

- What is the effect of stimulant treatment in autistic children on functional connectivity?
- ABIDE I/II school-age children: 27 autistic children taking stimulants, 101 not (ages 8–13).
- Treatment can also change motion: motion is a mediator.

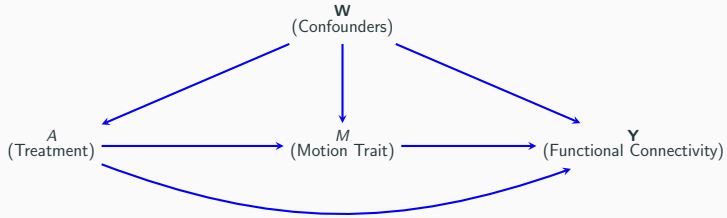
Causal diagram with derived outcomes



Causal mediation with derived outcomes in the presence of intra-subject nuisance sources of variance.

Red: latent variables.

Group-level DAG with marginalized latent variables



The group-level mediation diagram after marginalizing latent variables and time.

Parameters of interest and estimation

- Decompose the average treatment effect into the natural indirect effect through motion and the natural direct effect not mediated by motion:

$$\tau_{ATE} = \tau_{NIE} + \tau_{NDE}$$

- Augmented inverse-probability-weighted (AIPW) estimator; cross-fitting for all nuisance models.

What the theory requires I: group level (n subjects)

- **Multiply robust:** nuisances are the propensities $\hat{\pi}(a \mid \mathbf{W})$ and $\hat{\pi}(a \mid M, \mathbf{W})$, the outcome regression $\hat{\mu}(M, A, \mathbf{W}) = \hat{E}[\hat{Y}_j \mid M, A, \mathbf{W}]$, and the sequential regression $\hat{\xi}(\mathbf{W}) = \hat{E}[\hat{\mu}(M, a, \mathbf{W}) \mid A = a', \mathbf{W}]$; consistent if one of the following pairs is consistent: (i) both propensities; (ii) $\hat{\mu}$ and $\hat{\pi}(a \mid \mathbf{W})$; (iii) $\hat{\mu}$ and $\hat{\xi}$.
- **Rates:** $\sqrt{n}$-normality needs only *products* of errors to beat $n^{-1/2} \Rightarrow$ each nuisance may converge as slowly as $n^{-1/4}$: flexible machine learning, cross-fit across subjects.
- Given the derived-outcome bias conditions (next slide), the group level is standard mediation semiparametrics (Tchetgen Tchetgen and Shpitser, 2012), in a mediator-density-free form (Bayes' rule + sequential regression).
- **Simultaneous inference over J edges** (Gaussian approximation + multiplier bootstrap, FDP exceedance control) additionally requires *uniform* control of the derived-outcome bias: $\sqrt{n} \log J = o(T)$ (Chernozhukov et al., 2013; Qiu et al., 2023; Du et al., 2025).

What the theory requires II: time-series level (T volumes)

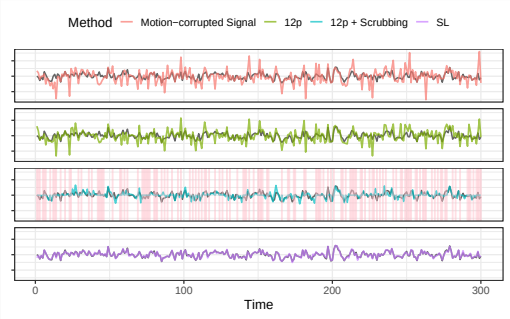
No robustness: biases enter at first order and must vanish faster than $n^{-1/2}$.

- **Processing bias** (learning motion artifacts; blocked cross-fitting in time): enters FC *quadratically* $\Rightarrow \sqrt{n}\, T^{-2\alpha_0} \to 0$, i.e. $T \gg n^{1/(4\alpha_0)}$, with $T^{-\alpha_0}$ the learner's prediction-error rate:
  - correctly specified parametric model (e.g., linear in motion parameters): $\alpha_0 = \frac{1}{2} \Rightarrow T \gg \sqrt{n}$.
  - additive, twice-differentiable components (MARS-type, additive splines): $\alpha_0 = \frac{2}{5} \Rightarrow T \gg n^{5/8}$.
  - random forests: slow worst-case guarantees.
  - SuperLearner tracks the best learner in its library.
- **Oracle bias** (result, under mixing): Fisher-z sample correlation is biased at $O(1/T)$; assuming $T \gg \sqrt{n}$ makes it negligible.
- **Temporal dependence:** theory assumes β-mixing: dependence between scans decays exponentially.

SuperLearner used for preprocessing, no scrubbing

- Mean model; generalized linear model.
- LASSO, ridge, elastic net (`glmnet`).
- Random forest (`ranger`).
- Multivariate adaptive regression splines (`earth`).

SuperLearner motion cleaning without censoring



Black: simulated motion-free signal. **Red**: motion-contaminated.

Green: 12p residuals. **Blue**: 12p + scrubbing. **Purple**: SuperLearner.

n	Method	Bias	Power	Type-I
150	12p+Linear X	0.10	0.99	0.56
	12p+Linear X+M	−0.19	0.25	0.22
	12p Scrub+Linear X	0.12	0.98	0.39
	12p Scrub+Linear X+M	−0.16	0.30	0.17
	SL+AIPW	**−0.04**	**0.49**	**0.06**
300	12p+Linear X	0.10	1.00	0.85
	12p+Linear X+M	−0.19	0.44	0.33
	12p Scrub+Linear X	0.12	1.00	0.60
	12p Scrub+Linear X+M	−0.16	0.52	0.27
	SL+AIPW	**−0.05**	**0.81**	**0.06**

1000 replications, $T = 300$. Bias and power for a region pair with true effect 0.3; type-I error for a null pair. SL+AIPW controls type-I error even at $n = 150$.

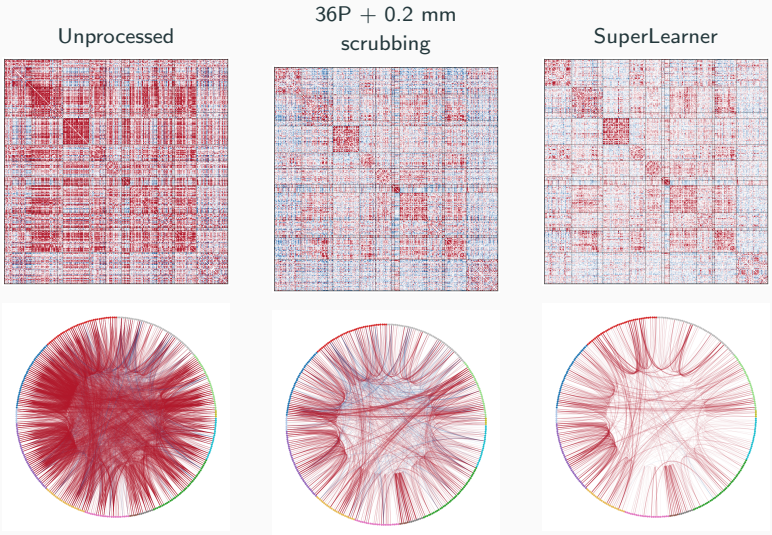
SuperLearner preprocessing: parcel time series

Same scan — nothing deleted.



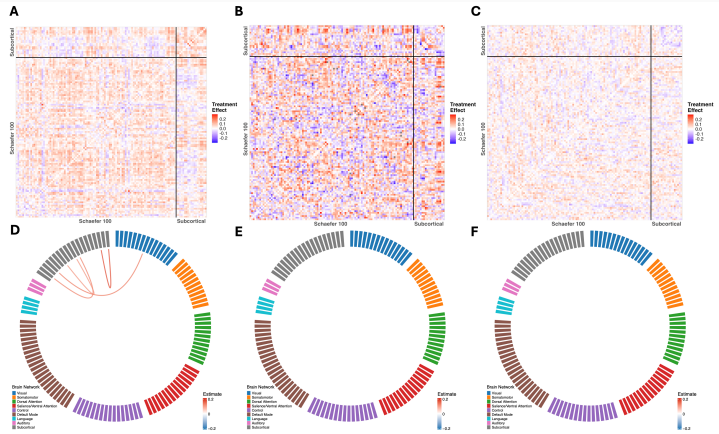
https://youtu.be/7jZPIX6s_Fk

Example single subject functional connectivity



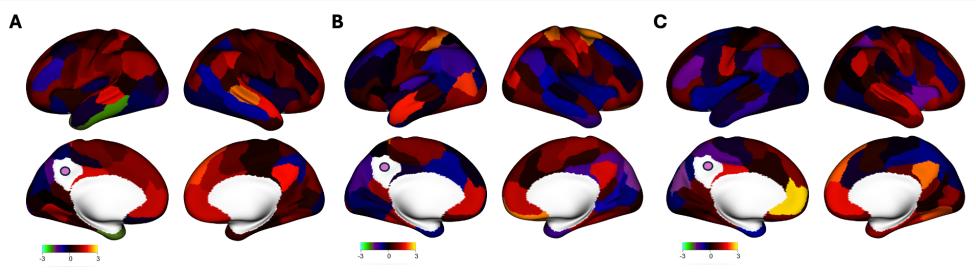
Top: Fisher-z correlation matrices (shared scale). Bottom: edges with $|z| \geq 0.5$; red positive, blue negative.

Group-level ABIDE results: stimulant NDE on connectivity



Stimulant impacts under "36p + Linear X" (A), "36p Scrub + Linear X+M" (B), and SL+SL (C); significant edges (FDR = 0.10 or $P(\mathrm{FDP} > 0.1) \leq 0.1$) in D–F.

Natural Direct Effects DMN seed-based approach



Stimulant impacts on FC between a posterior DMN seed and all other regions, estimated using "36p + Linear X" (A), "36p Scrub + Linear X+M" (B), and SL+SL (C).

Limitations

- Treatment confounding by indication: we do not have the symptoms prior to treatment.
- Small sample size.
- Children did not take stimulants in the 24 hours before the scan, so the analysis targets long-term changes.
- For fast acquisitions, may need to combine SuperLearner cleaning with lenient scrubbing of extreme volumes only.

Synthesis

Two solutions, two levels

	MoCo	**SL time-series cleaning**
Works on	processed FC	raw time series
Retains	all participants	all participants, all frames
Question	group differences under low motion	treatment effects with motion as mediator
Needs	any processed dataset	access to time series

- Both replace exclusion with estimation: use all participants, model the motion.

Future work

- **Connectivity analysis of the Emory Autism Brain Connectivity Study**: fast acquisition (1 second between volumes), 14 minutes, rich phenotyping.
- **MoCo beyond rs-fMRI**: the framework needs only a scalar quality variable — T1w morphometry with the Euler number; diffusion MRI with eddy-based RMS displacement (the analogue of mean FD).
- **Improve SuperLearner for time series**: real-data validation, time-series-aware learners, principled censoring of extreme volumes.

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The Emory Autism Brain Connectivity Study team

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